

Numerical Simulation of a Bending Test on a Carbon/Epoxy Laminated Composite Using ABAQUS

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Abstract

This work presents the finite element analysis and modeling of a bending test on a laminated composite specimen featuring an organic matrix (epoxy) reinforced with continuous carbon fibers. The primary objective is to simulate the mechanical behavior of a symmetric $[0/45/-45]_s$ layup subjected to a uniform surface pressure under simply supported boundary conditions on its short edges using ABAQUS/CAE.

The study focuses on evaluating the three-dimensional Von Mises stress distribution through the thickness of the laminate, the evolution of local stress and strain components $(\sigma_{11}, \epsilon_{11})$ and $(\sigma_{22}, \epsilon_{22})$ within the plies, and the mesh convergence of the numerical model. A comparative analysis regarding the impact of the alternative stacking sequence $[45/-45/90]_s$ and the constituent material properties (carbon/epoxy versus glass/epoxy) on the maximum deflection U_3 is conducted to highlight anisotropy effects, coupling mechanisms, and structural optimization specific to laminated thin plates.

Keywords: Composite materials, Laminates, Finite Element Method, ABAQUS, Bending, Orthotropy, Stacking sequence, Mesh convergence

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2. RESULTS AND PROJECT DISCUSSION

2.1. Question 1: Von Mises Stress Distribution Through the Thickness

The through-thickness distribution of the Von Mises stress exhibits a sharp, discontinuous jagged profile at the boundaries between adjacent layers. This phenomenon is a direct consequence of the sudden change in fiber orientation across ply interfaces (from 0° to $\pm 45^\circ$), which induces severe directional stiffness mismatches (jumps in elastic moduli). While the kinematic displacement and strain fields remain continuous across the perfectly bonded interfaces to maintain structural integrity, Hooke’s law dictates that the resulting stresses must adjust abruptly to compensate for these local stiffness variations, creating the observed stair-step stress profile.

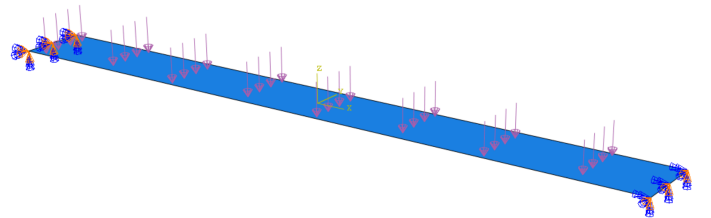


Figure 1. Boundaries conditions

1. INTRODUCTION: CONTEXT AND OBJECTIVES

Laminated composite materials are widely integrated into modern aeronautical and space structures due to their exceptional strength-to-weight and stiffness-to-weight ratios. Unlike conventional isotropic metallic materials, composites exhibit intrinsic heterogeneity and anisotropic behavior (typically orthotropic for a unidirectional ply) dictated by the orientation of the reinforcing fibers relative to the macro-level loading axes.

The analytical modeling of thin or moderately thick plate structures traditionally relies on Classical Laminate Theory (CLT) based on Love-Kirchhoff kinematic assumptions, or First-order Shear Deformation Theory (FSDT) like Mindlin-Reissner to account for transverse shear. However, complex thermomechanical gradients, susceptibility to interlaminar delamination, and complex elastic coupling mechanisms render the Finite Element Method (FEM) indispensable for the numerical validation of aerospace components.

In this project conducted on ABAQUS/CAE, a composite plate is modeled using shell continuum elements. The base specimen consists of a 6-ply symmetric carbon/epoxy layup denoted as $[0/45/-45]_s$, where each ply possesses an elementary thickness and orthotropic elastic properties defined by engineering constants $(E_1, E_2, \nu_{12}, G_{12})$. Analyzing the Von Mises stress state, localized strains, mesh discretization sensitivity, and geometric optimization of the stacking sequences constitutes the core of this engineering report.

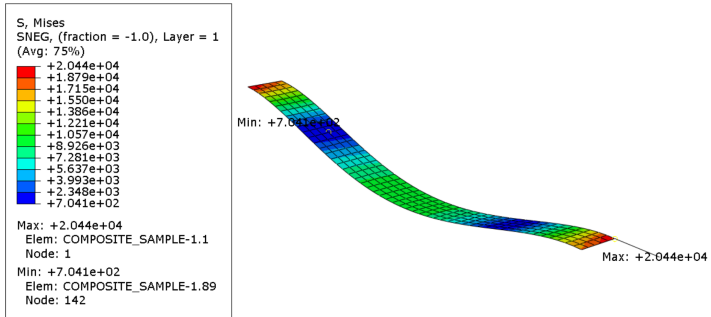


Figure 2. Stress distribution

2.2. Question 2: Critical Layer Identification

The VM stress reaches its maximum in the outermost layers (Layers 1 and 6 at 0°) due to the peak bending moment combined with the high longitudinal stiffness of the fibers, whereas it drops to a minimum at the mid-plane neutral axis (interfaces of Layers 3 and 4) where bending strains vanish.

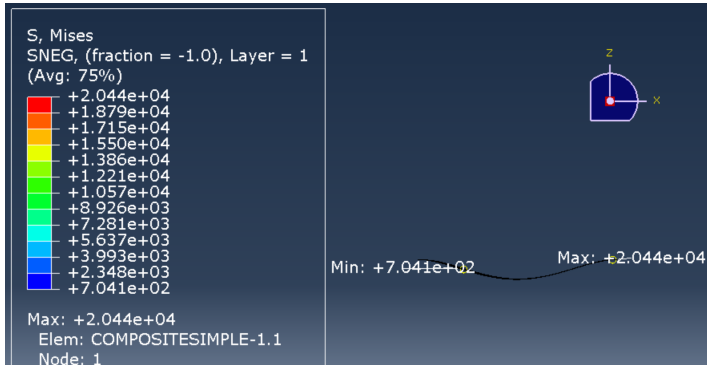


Figure 3. Max and min value of stress

2.3. Question 3: Global Specimen Contour Plot

See Figure 2.

The global contour plot of the Von Mises stress displays a symmetric distribution across the specimen, reaching its maximum absolute value of 105.7 MPa at the geometric center of the plate. This peak zone directly correlates with the location of the maximum bending moment induced by the uniform surface pressure. Conversely, the stress drops to its lowest values near the short edges, confirming the baseline state expected near the simple support boundary conditions.

2.4. Question 4: Local Material Response ($\sigma_{11}, \epsilon_{11}$)

The $(\sigma_{11}, \epsilon_{11})$ material response extracted from the central element in the longitudinal direction (X-axis) demonstrates a strictly linear elastic behavior up to the peak loading state. In contrast to the transverse response, this configuration exhibits an extremely steep slope (stiffness), which directly reflects the high longitudinal Young's modulus ($E_1 = 138,000$ MPa) of the unidirectional carbon fibers. Because the continuous fibers are perfectly aligned parallel to the primary bending normal stress path along the length of the specimen, they act as the main load-bearing constituents of the laminate. Furthermore, the negative values of both stress and strain confirm that this specific integration point is located within the upper plies of the specimen, operating under pure compressive bending conditions.

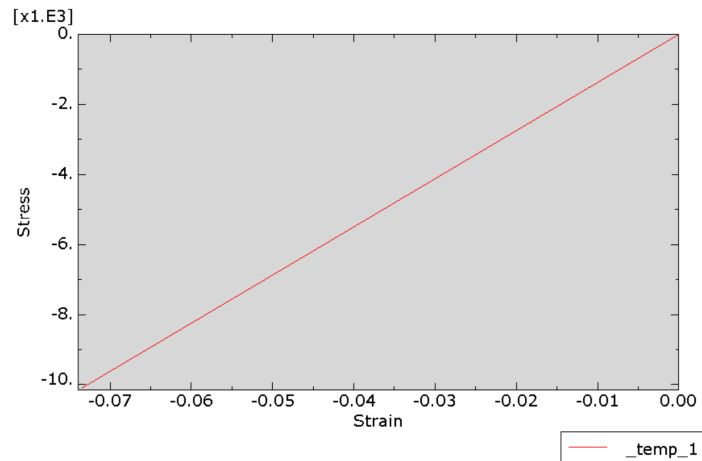


Figure 4. Stress-strain curve (σ_{11} vs ϵ_{11}) for the element in the middle.

2.5. Question 5: Local Material Response ($\sigma_{22}, \epsilon_{22}$)

The $(\sigma_{22}, \epsilon_{22})$ response extracted from the highest-stress element displays a perfectly linear elastic behavior operating in pure compression, where the negative stress and strain values confirm its location on the upper face of the specimen under bending.

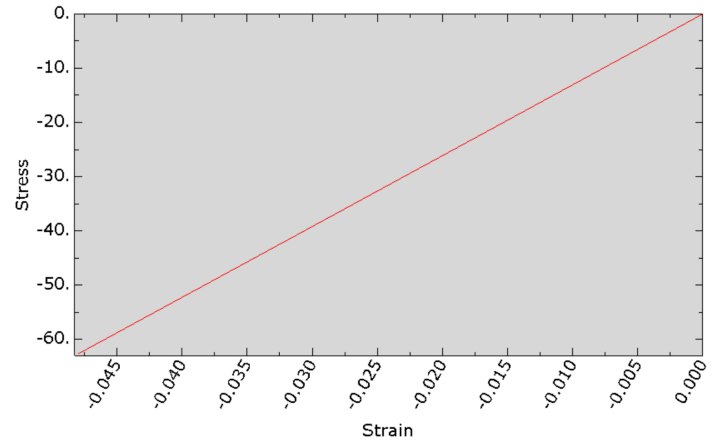


Figure 5. Stress strain curve for the element with max stress

2.6. Question 6: Analytical Homogenization Calculation

The engineering constants are verified by extracting the localized slopes from the finite element model. For the longitudinal direction along the fibers, the numerical calculation yields:

$$E_x = \frac{\Delta\sigma_{11}}{\Delta\epsilon_{11}} = \frac{10,206 \text{ MPa}}{0.0742} \approx 137,547.17 \text{ MPa} \approx 137.55 \text{ GPa}$$

Comparing this to the theoretical modulus ($E_1 = 135,000$ MPa) shows excellent agreement, with a relative error of only 1.89%, validating the shell formulation under bending.

Conversely, the direct uniaxial extraction in the transverse direction results in:

$$E_y = \frac{\Delta\sigma_{22}}{\Delta\epsilon_{22}} = \frac{63 \text{ MPa}}{0.048} \approx 1,312.5 \text{ MPa} \approx 1.31 \text{ GPa}$$

This reveals a major discrepancy compared to the isolated ply properties ($E_2 = 9,500$ MPa). This difference is normal and physically expected; a simple stress-strain ratio fails here because the sampled element is embedded inside a multi-directional $[0/45/-45]_s$ sequence.

An accurate evaluation requires advanced homogenization via Classical Laminate Theory (CLT) to account for structural stiffness rotations ($\bar{Q}_{ij}$) and Poisson's ratio couplings induced by the angled $\pm 45^\circ$ plies. Although this laminate methodology will be fully addressed next year, this mismatch directly illustrates the complex anisotropic behavior inherent to multi-layered composite structures.

2.7. Question 7: Mesh Sensitivity and Convergence Analysis

The mesh sensitivity analysis reveals a clear dependency between the spatial discretization and the captured stress levels. As the global mesh size decreases from 5.0 mm to 2.0 mm, the maximum recorded Von Mises stress increases progressively, rising from its lowest value of approximately 18,000 to nearly 20,000 at a size of 2.5 mm, before reaching its peak at about 20,400 with a 2.0 mm mesh. This upward trend demonstrates that a smaller, more refined mesh is crucial to properly capture localized stress gradients and minimize discretization errors.

In terms of convergence, the solution shows distinct stabilization behavior between the 2.5 mm and 2.0 mm configurations, where the slope of the curve flattens significantly compared to the sharp initial rise from 5.0 mm to 2.5 mm. Although the stress value is still slightly increasing—indicating that strict asymptotic convergence has not yet

been fully reached—the curve heavily trends toward its limit. Consequently, a mesh size of 2.0 mm is chosen as the optimal selection for the final model, providing an excellent engineering trade-off by ensuring highly accurate stress resolution while maintaining low computational costs.

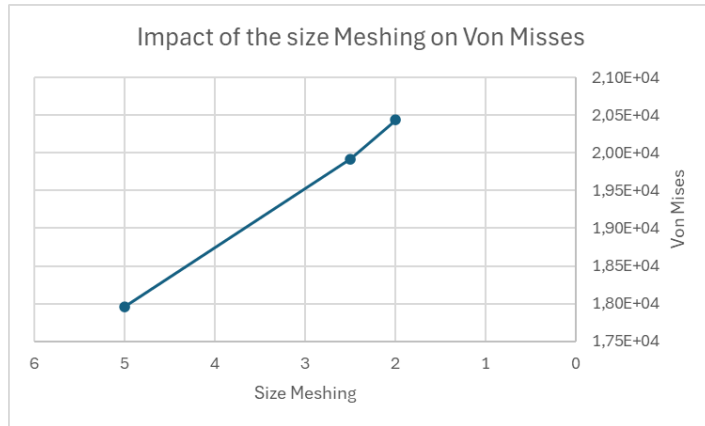


Figure 6. Maximum Von Mises Stress Sensitivity with Mesh Sizing Curve

Note

Element Technology (S4R): The selection of the S4R element (a 4-node, doubly curved thin/thick shell with reduced integration and hourglass control) is key to the model's reliability. First-order full-integration elements often suffer from "shear locking" in bending scenarios, overestimating stiffness. Reduced integration completely bypasses this numerical error, while the built-in hourglass control prevents spurious zero-energy modes, providing high structural fidelity at a low computational cost.

2.8. Question 8: Influence of the Stacking Sequence on Out-of-Plane Deflection U_3

We first plot the U_3 deformation of the initial stacking layout:

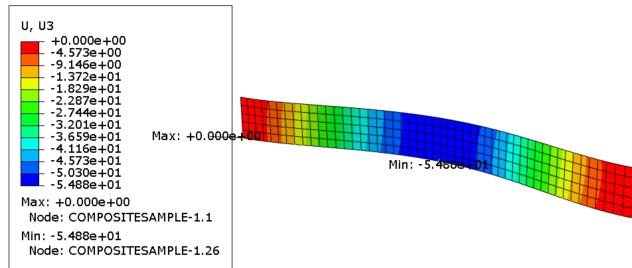


Figure 7. U_3 shape deformation with initial stacking layout

To analyse the effect of the stacking sequence on the carbon/epoxy material, we compare the initial simulation with the $[45/-45/90]_s$ laminate.

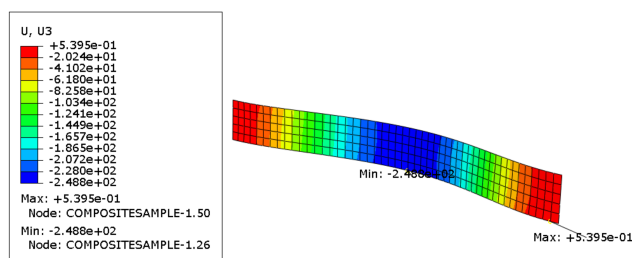


Figure 8. U_3 shape deformation with a different Stacking Sequence

Altering the initial stacking sequence to a $[45/-45/90]_s$ laminate increases the maximum U_3 deflection from -54.88 mm to -248.80 mm.

This significant loss in flexural stiffness indicates that the initial sequence likely contained 0° plies aligned with the bending axis; their removal severely reduces the D_{11} stiffness term.

Furthermore, while the initial sequence exhibits pure symmetric bending, the $[45/-45/90]_s$ laminate displays bending-twisting coupling, evidenced by an asymmetric contour and a positive U_3 displacement of $+0.5395$ mm. Although the laminate is symmetric (meaning the B matrix is zero), positioning the $+45^\circ$ and -45° plies at different distances from the mid-plane creates non-zero D_{16} and D_{26} terms, driving the observed twist.

2.9. Question 9: Influence of Fiber Type (Carbon vs Glass)

We first plot the U_3 deformation of the initial stacking layout:

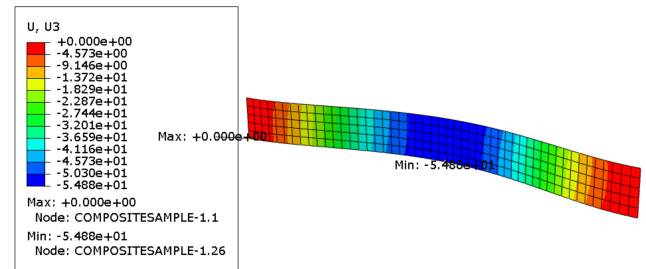


Figure 9. U_3 shape deformation with initial stacking layout

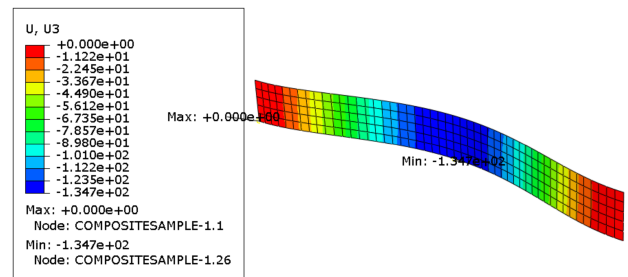


Figure 10. U_3 shape deformation with Glass Epoxy

Comparing the materials using the initial stacking sequence reveals that both composites maintain pure symmetric bending ($D_{16} = 0$). However, the glass/epoxy composite reaches a maximum deflection of -134.70 mm, deflecting nearly 2.45 times more than the carbon/epoxy composite (-54.88 mm).

This difference is governed entirely by the fiber properties. Carbon fibers possess a considerably higher longitudinal Young's modulus (E_1) than glass fibers. Consequently, the carbon/epoxy composite has a much higher global bending stiffness (D_{11}), allowing it to effectively resist deformation under identical loading conditions.

3. CONCLUSION

This finite element simulation work using ABAQUS/CAE validated the governing constitutive behavior of orthotropic laminated plates subjected to transverse bending loads. The analysis highlighted the discontinuous stress distribution through the thickness at ply boundaries resulting from stiffness jumps, alongside the fundamental requirement of mesh convergence verifications to ensure finite element answer reliability.

The optimization steps conducted on the stacking sequence demonstrated that adjusting the ply layup pattern relative to critical stress paths (retaining 0° layers for longitudinal bending) combined with opting for high specific performance constituents (carbon/epoxy over glass/epoxy) remains vital to contain structural out-of-plane U_3 displacements within strict aerospace design tolerances.